

MATH – 2307**Group -A : 20 Marks**

1 – Definition. LU D-Composition

– **10 Mark****Vector & Scaler :****Defi Vector & Scaler**

From Book - 24,29

Dot & Cross :

Defi Dot & Cross Product.

From Book – 13,16

Group – B : 30**Vector d/dx :** Book - 8,12,16**line Integral** – pdf 7,9,16.**Surface Integral pdf****- 10 Mark**

➤ Define Green's Theorem &

Q# 87: Evaluate the line Integral $I = \oint_C \{xydx + (2x - y)dy\}$ round the region bounded by the curves $y = x^2$ and $x = y^2$ by the use of Green's theorem.

******* Verify the Green's theorem in the plane for $\int \{xy + x^2\} dx + xy^2 dy\}$ where c is the close curve of the region bounded by $y = x$ and $y^2 = x$.

or

3 - Volume Intregral - pdf**10 Mark Only Variable Change.**

Q # 77: Verify the divergence theorem for the vector field $\vec{F} = x^2 \hat{i} + z \hat{j} + y \hat{k}$ taken over the region bounded by the planes $z = 0, z = 2, x = 0, x = 1, y = 0, y = 3$

For Practise Q – 75 , 76 , 78.

This Will Cover 60-70% for final preparation. For More Follow DMEK Sir Suggestion .

MATRIX

EIGEN DECOMPOSITION, LU DECOMPOSITION, SINGULAR VALUE DECOMPOSITION (STATEMENT)

EX-67, 68, 70

VECTOR ANALYSIS

STATEMENT: DOT PRODUCT, CROSS PRODUCT, PROJECTION OF A VECTOR, UNIT VECTOR, CO-PLANER, COLLINIAR, NORMAL VECTOR, SCALAR FIELD, VECTOR FIELD, GRADIENT, DIVERGENCE, CURL, SOLENOIDAL, IRROTATIONAL VECTOR FIELD, CONSERVATIVE VECTOR FIELD, DIRECTIONAL DERIVATIVE, SINK FIELD, DIVERGENCE THEOREM, GREENS THEOREM

EX-13,14,15, 17,18, 19,22,23, 25, 27, 28, 32,33, 35, 37, 38, 43, 44, 46, 56, 58, 61, 74, 76, 80, 84, 90, 91, 95, 99, 100, 110, 111